13 Here are the first six terms of a quadratic sequence.

10 19 34 55 82 115

Find an expression, in terms of n, for the nth term of this sequence.

First differences: 9, 15, 21, 27, 33

Second differences: 6, 6, 6, 6

1 mark
Constant second difference.

$6 \div 2 = 3$, so start with $3n^2$.

1 mark
Half gives coefficient of n^2.

$3n^2$ gives 3, 12, 27, 48, 75, 108.

Each original term is 7 more: nth term = $3n^2 + 7$.

1 mark
Add the remaining 7.

(Total for Question 13 is 3 marks)

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